UnionLabs: new agent, features, physical layers

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Institution

ACM MobiCom ’26 Demo · Austin, TX · October 2026

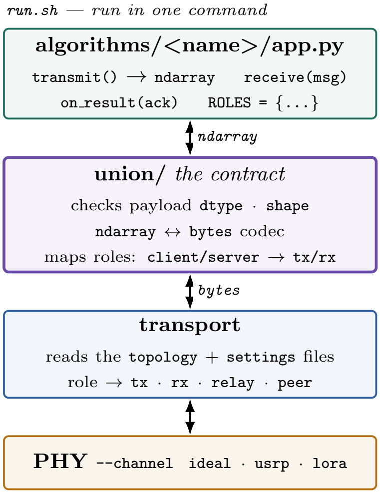
The problem: everything around the algorithm

- A learning-over-radio experiment is easy to state and expensive to run — the algorithm is a page of code; what surrounds it is not.
- Platforms removed the hardware barrier, open stacks the waveform barrier ...
- ... but users still hand-link algorithms to the physical layer: hundreds of lines, days of debugging.
- The wiring lives in operators' heads and shell history — retyped per machine, and the retypings disagree.

UnionLabs separates what an algorithm does from how the experiment is wired.

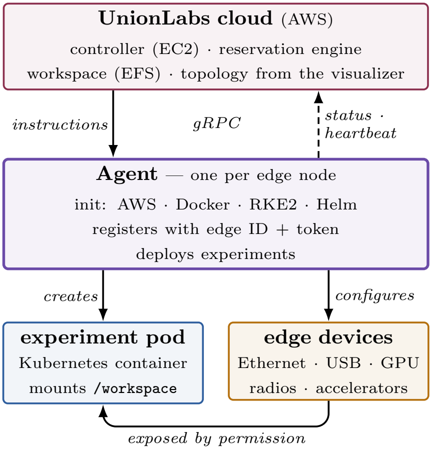
Four designed parts

- **The Agent** — any machine becomes a managed edge node
- **The shared workspace** — one account's state, across testbeds
- **The algorithm contract** — `transmit()` / `receive(msg)`, nothing else
- **The PHY codes** — a modem whose every stage is selectable



The UnionLabs Agent

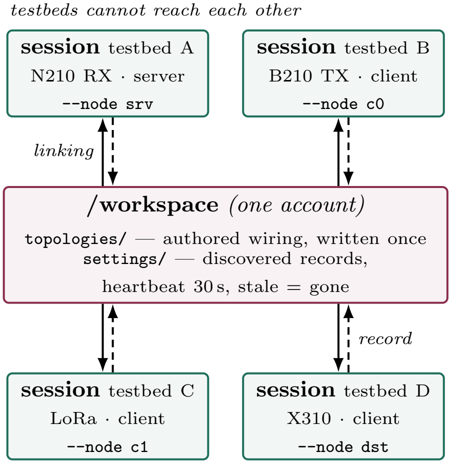
- One agent per edge node — server, workstation, or laptop.
- Install: prepares AWS, Docker, RKE2, Helm; registers with a unique edge ID + token.
- gRPC with the cloud: instructions down, status + heartbeats up.
- Owner uploads the topology via the visualizer; reservation allocates from it — live nodes only.
- Experiments become Kubernetes pods; devices exposed *only by the owner's permission*.
- Remote access rides the cloud–edge path: pods never publicly exposed.



Remote access rides the same cloud–edge path:
pods are never publicly exposed.

One workspace across testbeds

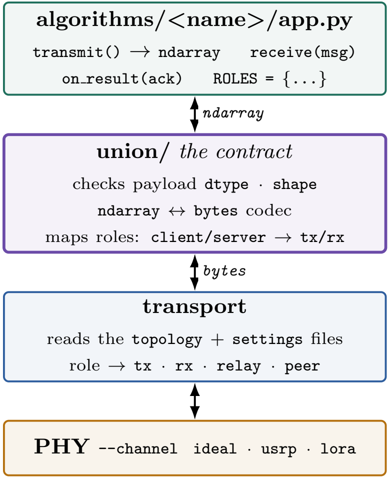
- Every session of an account mounts the same `/workspace` — even when the testbeds cannot reach each other.
- `topologies/`: authored wiring, written once.
- `settings/`: each session's own record — 30 s heartbeat, dropped when stale.
- Two kinds of state, two lifecycles: the shared file system never carries a fact that can go stale.



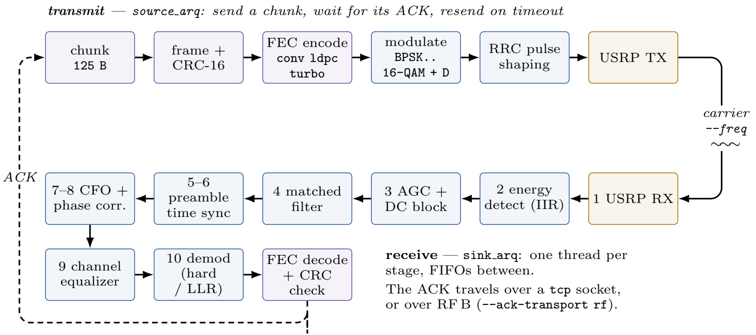
The algorithm contract — `run.sh`, one command

- The algorithm writes `transmit()` and `receive(msg)` — and imports nothing from any radio driver.
- The union layer validates payloads, converts `ndarray` ↔ bytes, maps role names.
- The transport reads the topology + settings files; the node becomes `tx` · `rx` · `relay` · `peer`.
- `--channel` selects the PHY: `ideal` · `usrp` · `lora`.
- Same path, both directions — change the physical layer with one flag, no code change.

run.sh — run in one command



The PHY codes



Waveform switch (sc → ofdm): RX stages 4–9 become one OFDM demodulator (FFT, CP removal, per-subcarrier equalization) and TX skips the RRC shaper. Same framing, same ARQ, same flags either side. This chain is the USRP modem; the same contract also rides the LoRa PHY and an ideal channel.

ARQ framing / coding DSP stage UHD / radio

- Modulation, coding, synchronization, waveform, gains, carrier — all selectable per experiment.
- Stop-and-wait ARQ end to end; the ACK over TCP or a second RF path.

Wiring as a file

- A topology names each node, its role, its radio, its ports, and the links between nodes — every node launches from the same file.
- **Per-link, per-direction media**: one hop over the air, the next over Ethernet; the reply may return over TCP/IP.
- A wiring that cannot run is **refused when it is read**, naming the node at fault — not minutes later, from the wrong layer.
- One command walks 44 flags and 82 topology settings from typed text to the object each configures.

Demonstration: each part, working

1. **Onboarding (Agent)** — a machine enrolled on the spot; heartbeat appears; the visitor opens the container's remote desktop.
2. **One workspace** — edit a wiring file here, see it from a session on a remote testbed.
3. **One algorithm, different PHYs** — radio-free, then real radios by one flag; delivery, retransmissions, airtime on screen.
4. **Rewiring, and refusal** — a multi-hop experiment changes medium by one field; an impossible wiring is refused by name.

Takeaway

Write the experiment once.
Run it over any PHY, topology, and testbed.

`./run.sh --algo <yours> --topology <wiring> --node <me>`

One command · one wiring file · one contract

Come run it at the table.